

Problem:-

1) A channel has a bit rate of 4 Kbps and one-way propagation delay of 30 msec. The channel uses stop and wait protocol. The transmission time of the acknowledgment frame is negligible. To get a channel efficiency of at least 75%, min frame size should be

Given

- Bandwidth (B) = 4 kbps = 4000 bps.
- propagation delay (T_p) = 30 msec
- Efficiency (η) = 75% or 0.75

* Solve for Transmission Time (T_t).

$$\text{Efficiency formula } \eta = \frac{T_t}{T_t + 2T_p}$$

$$0.75 = \frac{T_t}{T_t + 2(30)}$$

$$0.75(T_t + 60) = T_t$$

$$0.75T_t + 45 = T_t$$

$$45 = T_t - 0.75T_t$$

$$45 = 0.25T_t$$

$$45 = 0.25T_t$$

$$T_t = 45 / 0.25$$

$$= 180 \text{ msec}$$

$$= 0.18 \text{ sec}$$

Calculate frame size (L)

$$L = T \times B$$

$$= 0.18 \text{ sec} \times 4000 \text{ bps}$$

$$= 720 \text{ bits}$$

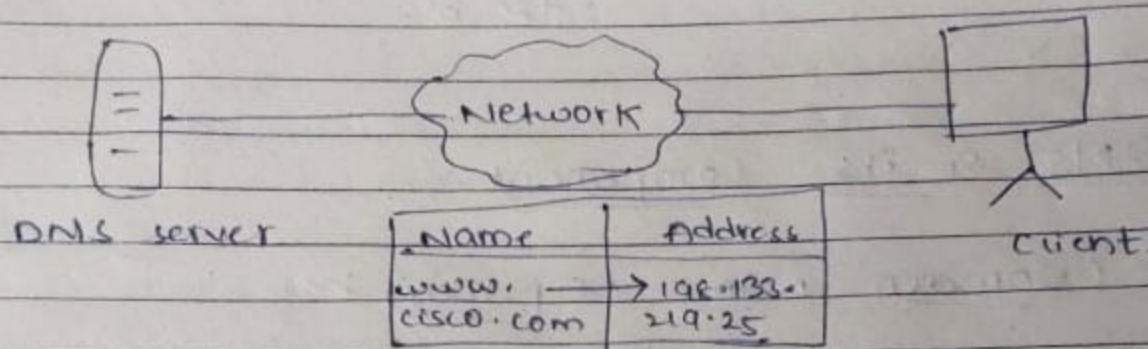
2) DNS & its components

DNS (Domain Name System) :-

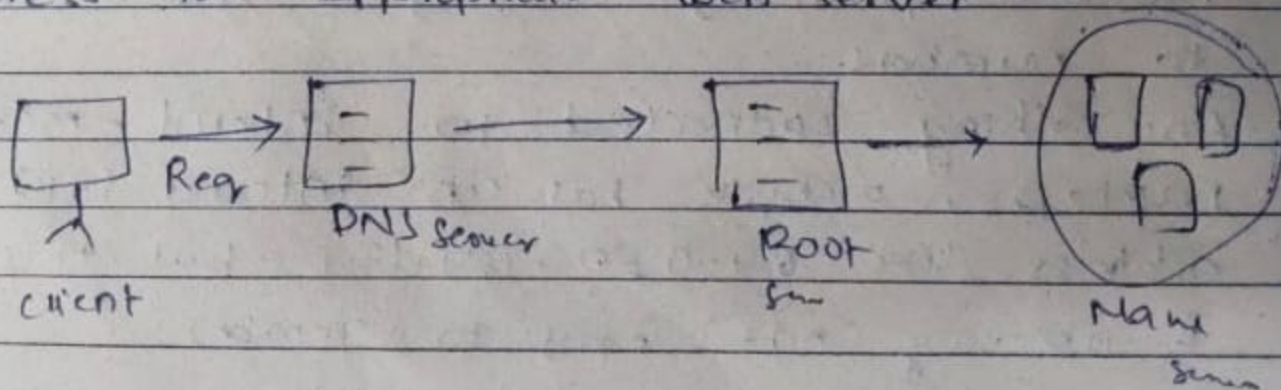
- An internet service that translates domain name into IP addresses. DNS resolver is a ~~resolver~~ server responsible for receiving a DNS query from a client and finding the corresponding IP address of a domain name. The domain name system (DNS) is a central part of Network providing a way to match names to numbers.
- Any thing connected to internet, mobile, laptops, websites has an internet (IP) address like 64.020.189.179. but this is obviously not easy to remember.

DNS Services & protocol

Resolving DNS Address



DNS server resolver names to IP address. It would be difficult to remember the IP address of every website we like to visit, but easy to remember names. When we enter a URL into our web browser, our DNS server uses its resources to resolve the name into IP address for appropriate web server.



Domain name Architecture:-

Domain name system comprises of Domain name, Domain name space.

Domain name Meaning:

- .com - Commercial business
- .edu - education
- .gov - government agency
- .int - international entity
- .mil - military
- .net - Networking organization
- .org - non-profit

Components:-

The domain name system consists of several components work together to enable name into IP address.

DNS Client:- The DNS clients send queries to DNS resolver to obtain IP address corresponding to a given name.

DNS Resolver:- First contact by client and takes a domain query from the client and looks for IP address. performs recursive lookup if needed. stores results in temporary.

Root DNS Server:- Top lvl of hierarchy. does store exact domain records. Direct TLD server based on domain extension (.com, .in, .net).

TLD Server (Top - lvl - domain server):- responsible for specific domain extensions like (.com, .org, .in).
• Direct resolver to the authoritative DNS.

4) Authoritative DNS server:-

- The final server that contains actual DNS records of a domain, provides the real IP address to the resolver. (ex:- server owned by cloudflare @ Go daddy).

5) DNS Records:- These stored in authoritative server includes A, AAAA, CNAME, MX, NS

6) DNS cache:-

- Temporary storage on client system.
- Speed up future lookups.
- Reduce loads on DNS infrastructure.

2) IP Addressing:-

- * IP address is mean that the internet protocol address
- * It is an unique address assigned to each computing device in IP network
- * ISP assigns IP address to all devices present on internet its network.
- * computing devices use IP address to identify and communicate with other devices in the IP network.

Types of IP Address:-

Static IP address:-

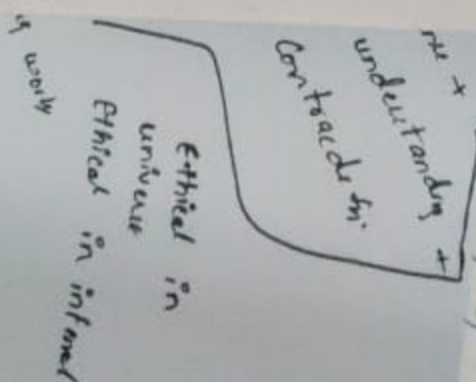
- * static IP address is an ip address that once assigned to a network element always it will be same.
- * they are configured manually
- * some ISP doesn't provide static ip add
- * it is higher costly than dynamic IP add.

Dynamic IP address:-

- * dynamic IP address is a temporarily assigned IP add to a network element. ↑ Hard to change
- * easy to manage.
- * provide by DHCP.

cheap

how



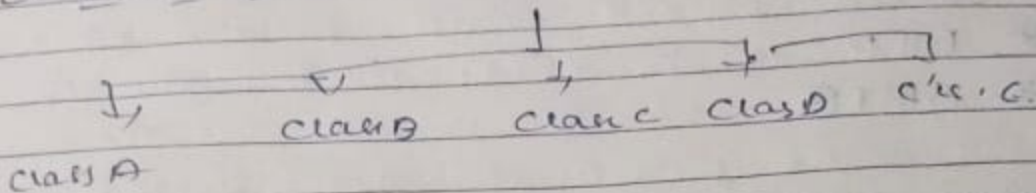
format of IP Address

Classification

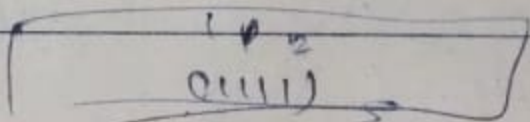
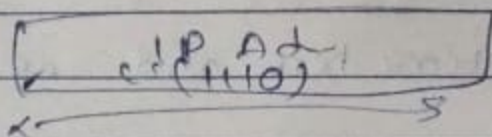
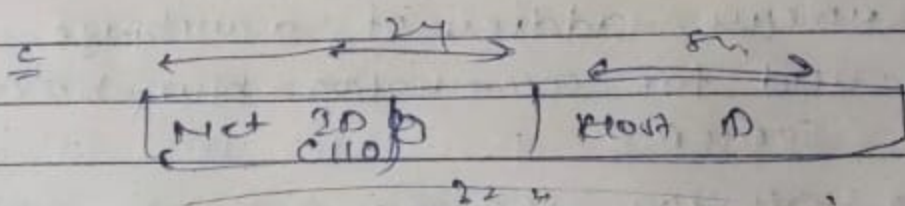
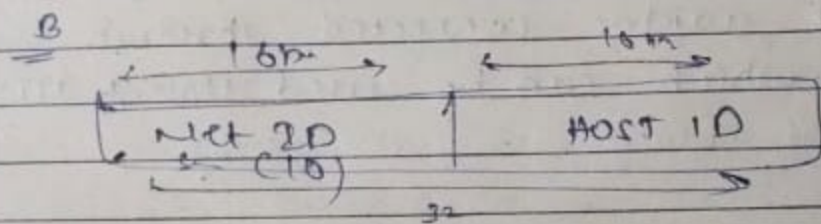
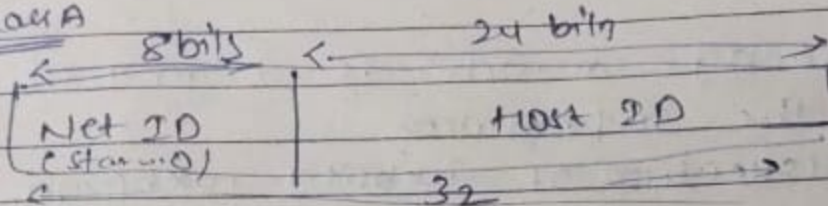
* classful Address system

↓
classless addressing system

classful address sys



class A



Why

ethical in

universe

ethical in internet

Contract in

Existence + Right understanding +

#

IP

Add

44 Overview summary :-

- * WWW
- * HTTP
- * RTP
- * RTSP
- * SNMP
- * FTP
- * SMTP

*7 www (world wide web) :-

- * It is simply a web is a way of accessing the information.
- * It is collection of websites, webpages, videos and audios connected through hyper links which can be used through HTTP components of www.

URL → unique address of a webpage

HTTP → used for communication b/w browser & Server

HTML → Display webpage

- * It supports multimedia and enables e-commerce, education

Applications:-

- Information search
- social networking
- online shopping

Ex:-

web browser

Google chrome

protocol

HTTP

URL

www.opnaticom

URL

protocol

* HTTP (Hyper text transfer protocol):-

- HTTP is a communication protocol used b/w a web browser (client) and a web server to transfer data on the world wide web.
- It is back bone of www.
- when client to retrieve a document on the internet web browser sends req to server & waits for reply then HTTP service handles that and by using TCP/IP sent document to client (Transmission process / in hand)

Applications:-

- 1) Browsing website
 - 2) online shopping
- or $\text{http} \parallel \text{www.hk.com}$

ethical in universe
ethical in internet
not using words

Right to understand +
Right to access in
Right to

IP Addressing

RTTP:-

- Real-time transfer protocol is a network protocol designed for delivering audio and video over the IP network.
- It is widely used in multimedia communication such as video calls, online.
- RTP is a packet based protocol, which means that it breaks the media stream into packets for transmission over the network. Each packet is given a sequence number, which allows the receiver to reassemble the packets into continuous audio/visual stream.
- RTCP (control) monitors Quality.

Ex:-

Zoom meet

Google meet

WhatsApp call

YouTube

RTSP (Real time streaming protocol)

RTSP is a network protocol used to control real time audio & video streaming. It supports commands like play/pause, stop and work together with RTP to deliver media data.

Ex:-

- CCTV / IP camera system
- video surveillance

- Video surveillance system
- VLC media storage

SNMP :- (Simple network management protocol)

- It is a popular protocol for network management.
- It is used for collecting information from servers, printers, hubs, switches on IP network
- SNMP is ~~used to~~ designed and used as request / response protocol. So the protocol details are simple (snmp)

components

- 1) manager / proxy
- 2) snmp agent
- 3) 4 manager
- 4) m2b

It has three versions
 snmp v1
 v2
 v3

Eg:-

cisco & juniper routers
 HP / DELL servers
 UPS units

ethical universe
 ethical in
 right contacts in
 co-existence +
 understanding
 each other

FTP (file transfer protocol)

- It is used to transfer files b/w client and the server over TCP/IP. We can upload document, data, images.
- It is one of the oldest & widely used protocol.

Components:-

- FTP client
- FTP server

App:-

- web hosting & maintenance
- ^{share} software updates

Ex:-

filezilla

SMTP (simple mail transfer protocol):-

- It is a standard application layer protocol used for sending mails from computer to another over the network.
- SMTP is responsible for
 - sending mails
 - forwarding mails
- It works mainly over TCP port 25.

AP:-

- Sending Email
- Server to server email transfer

G:-

smtp.gmail.com

5 - your

Q7 Data link layer :-

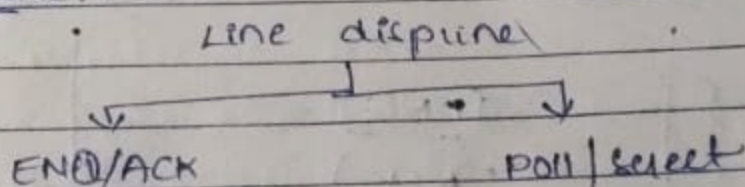
It helps two directly connected devices send data safely without errors. packs the data into small pieces called frames. Adds sender & receiver address called MAC address.

It is 2nd layer of OSI model

Application
Presentation
Session
Transport
Network
Data link
Physical

Line discipline (who should send now?)
Flow control (How much data may sent?)
Error control (How can errors may corrected?)

Line discipline :-



- Line discipline is a function of Data link layer.
- It makes sure that two devices communicate in a organized way.
 - One device sends
 - The other receives
 - NO confusion, NO overlapping msg

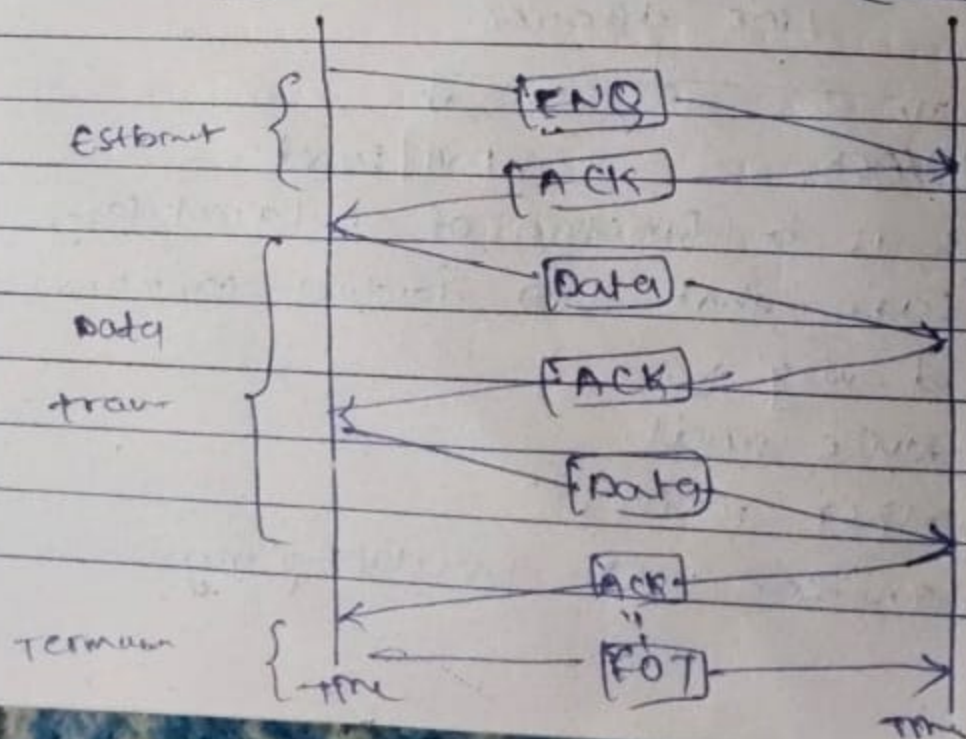
• ENQ / ACK :-

• Enquiry / Acknowledgement

- The sender first sends an ENQ signal check to the whether receiver is ready to receive data.
- If the receiver is ready, it replies with an ACK signal.
- After receiving the ACK, the sender transmits the actual data.
- Finally, the receiver sends another ACK to confirm that the data has been received correctly.
- This method ensures organized, error-free and Co-ordinated communication b/w two devices.
- Used b/w two devices

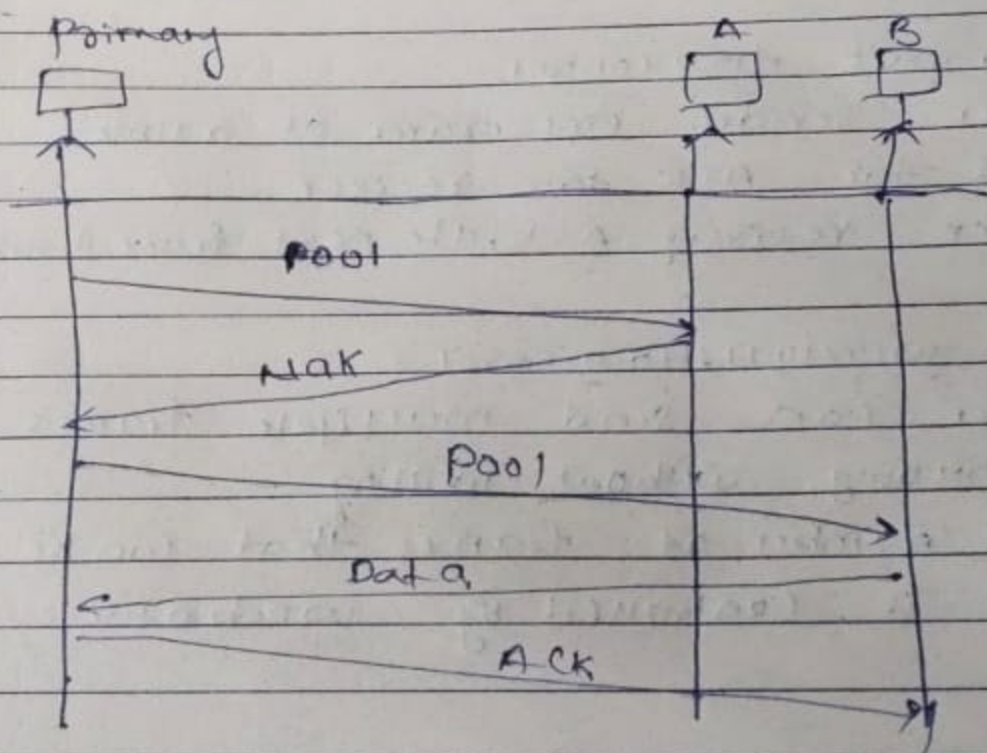
Host A

Host B



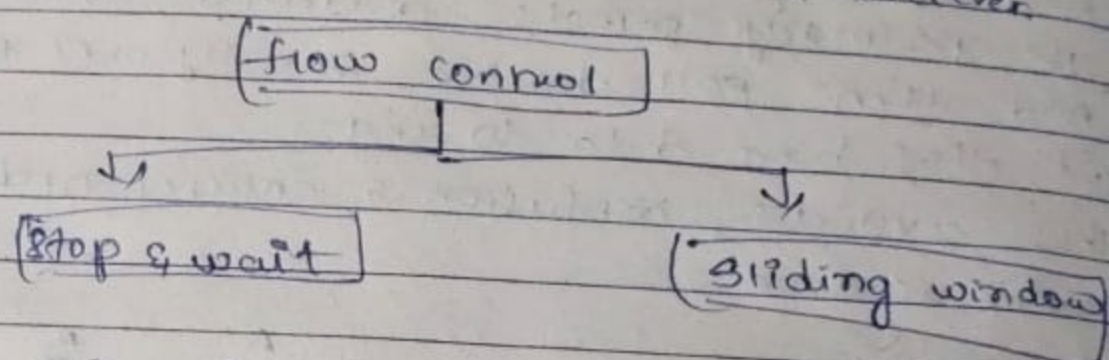
Poll / select.

- ↳ Polling / selection is used when multiple devices share a one communication line.
- ↳ There are only one primary device.
- ↳ And many secondary devices.
- ↳ The primary selects a device to receive data.
- ↳ And then poll devices one by one to check if they have data to send.
- ↳ This avoids confusion & ensures orderly communication.



Flow Control:-

- flow control ensures is a technique used in data link layer of computer networks to control the rate of data transmission b/w a sender & Receiver.



stop & wait flow control:-

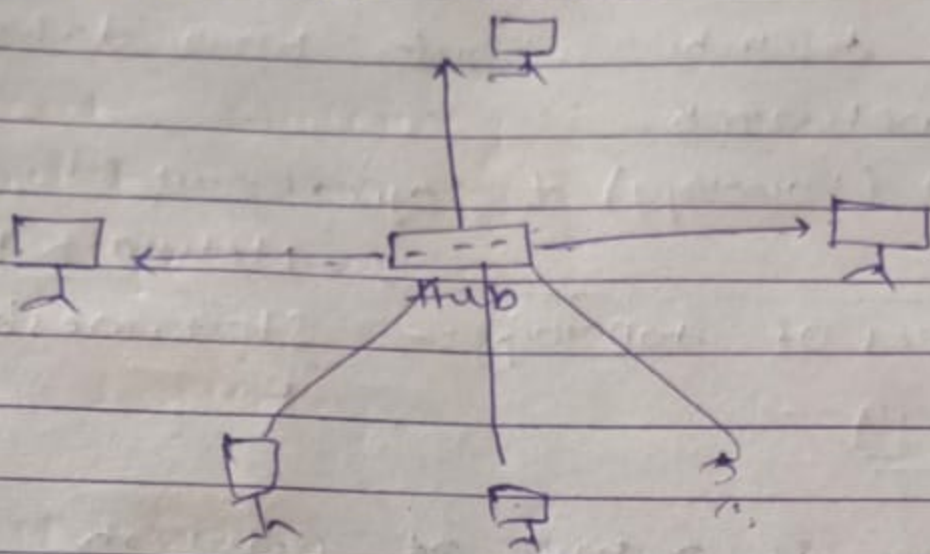
- sender sends one frame at a time
- waits for ACK from receiver
- After receiving ACK, the next frame is sent

Sliding window flow control:-

- sender can send multiple frames continuously without waiting
- The number of frames that can be sent is controlled by window size.

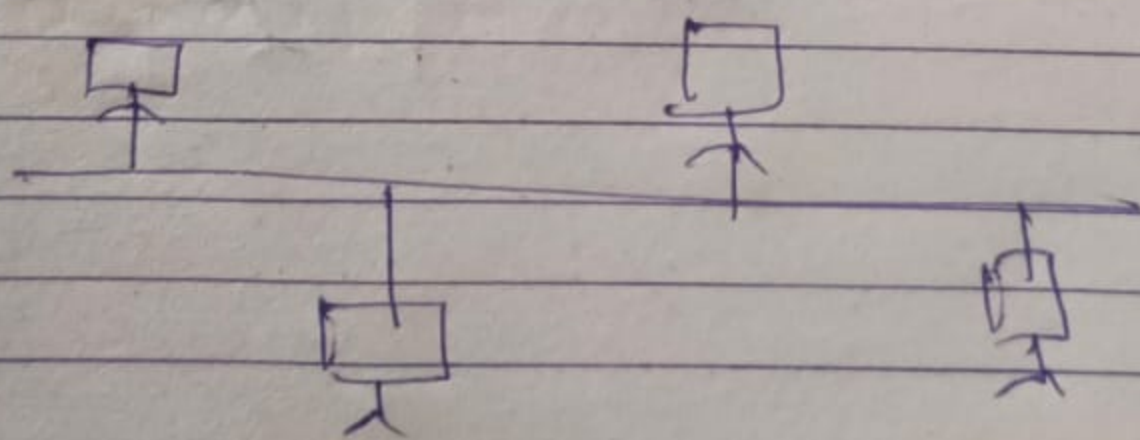
Star Topology:-

A star topology is one of the most common network setups. All devices (computers or nodes) are connected to a central device like a hub, switch or Router. The central network device acts as a server and other devices act as client.



Bus Topology

Bus topology is a network setup where all devices are connected to a single, central cable called bus. The cable is also known as back bone cable.



trust.

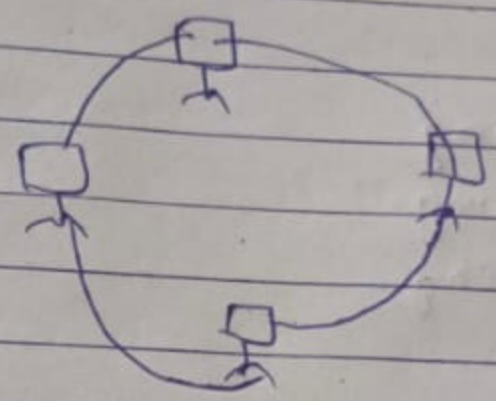
engng

lsh

Page Date

Ring Topology :-

Ring topology is a network configuration where devices are connected in a circular, closed-loop path, with each device is linked to exactly two other devices. Data travels in one direction around the ring.



Switching techniques :-

Switching technique is a method used in computer networks to transfer data from source to destination through intermediate devices (like switches, routers).

3 main types

- 1) Circuit switching
- 2) packet switching
- 3) message switching.

Circuit switching :-

Circuit switching technique is a switching technique. A dedicated communication path is established b/w receiver & sender before data transfer begins.

Phases :-

- > Establishing a circuit
- > Transferring the data
- > Disconnecting the circuit. (Telephone)

Packet Switching is a communication technique where the message are divided into small data units called packets. each packet travels independently through the network may take different routes to reach each destination. At receiver side all packets are reassembled in correct order. (Network)

- Data p.
Datagram ps:- each packet is independent. no fixed path. Order may change.
- Virtual circuit ps:- path is decided before sending packets. but no dedicated line like circuit switching.

message switching:- (Telegraph)

- store and forward technique are used to transfer data from source to destination. entire msg is treated as a single unit and stored and forwarded from node to node.
- Telegraph

